

MATERIAL SAFETY DATA SHEET

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1. IDENTIFICATION OF THE SUBSTANCE/ PREPARATION AND COMPANY / UNDER TAKING	
PRODUCT NAME : 2,4-D DIMETHYL AMMONIUM 84% W/V SL	
Address	: Atul Limited – Agrochemicals Division. Post : Atul 396 020 Dist : Valsad Gujarat state INDIA
Phone No.	: (91 - 02632) 233261, 233331
Fax No.	: (91 – 02632) 233024 233619
2. PRODUCT INFORMATION	
The product contains active ingredients : 2,4-D ACID	
Product Code :	407010 UN : 3082
HAZARDOUS :	Yes CAS No : 2008-39-1 Symbol : X _n R-22 R-20 / 22
Hazardous Ingredints : 2,4-D ACID	
3. HAZARDS IDENTIFICATION	
This product unlikely to cause harmful effects under nomal conditions of handling and use. Can cause harmful effects if not handled and stored as per requirment	
4 FIRST –AID MEASURES	
Inhalation	: Remove patient from exposure . Obtain medical attention if ill effects occur.
Skin	: Remove contaminated clothing. Wash skin with water . If symptoms develop .obtain medical attentoin . Contaminated clothing should be laundered before re –issue.
Eye Contact	: Irrigate promptly with plenty of water. If problem persists obtain medical attention.
Ingestion	: If swallowed induce vomiting by giving a tablespoon of salt in a glass of warm water. Repeat until vomit fluid is clear. If problem persists obtain medical attention.
Antidot / Treatment : Gasteric lavage , Quinidine sulphate to counter myotonia and ventricular fibrillation, supportive and symptomatic treatment.	

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5. FIRE FIGHTING MEASURE The material may form flammable clouds in air. Extinguishing media : Water spray, foam, dry powder or carbon dioxide. Fire protective equipment : A self contained breathing apparatus and suitable protective clothing should be worn if involved in fire with other materials.
6, ACCIDENTAL RELEASE MEASURES Protect against dust. Ensure suitable personal protection during removal of spillages. Clear up spillages and transfer to container for disposal. Wash the spillage area again.
7, HANDLING AND STORAGE 7.1 HANDLING Control dust formation. Avoid contact with skin and eyes. Avoid inhalation of high concentration of dust. Take precautionary measure against static discharge. 7.2 STORAGE Keep container tightly closed away from light and humidity. Keep away from reach of children.
8. EXPOSURE CONTROLS / PERSONAL PROTECTION Wear protective equipment to comply with good occupational hygiene practice. Respirators : Wear suitable respiratory protective equipment exposure to high levels of dust are likely. Skin protection : Use of apron is recommended to avoid skin contact. Eye protection : The use of goggles is recommended.
OCCUPATIONAL EXPOSURE LIMITS Hazardous Exposure Limits In Air : 1. TWA is 10 mg/m³ On daily average basis. 2. STEL is 20 mg/m³ On daily average basis.

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9. PHYSICAL AND CHEMICAL PROPERTIES	
Form	: Liquid
Colour	: Brown colour liquid.
Odour	: Aliphatic amine
pH (Value)	: 8 to 9 10 gm/lit water
Boiling point(°C)	: Not applicable
Meltig point (°C)	: Not applicable
Flash point	: Combustible liquid
Flammable limits	: Not pertient
Auto ignition temp (°C)	: Not pertient
Explosive Properties	: See ' Fire Fighting Measures'
Oxidising properties	: Strong oxidizers
Vapour pressure (mm Hg)	: 0.4 mm @ 320°C (- 1.1°C)
Density (gm/ml)	: 1.2 to 1.24
Soubility in water	: Miscible
Partition coefficient	: Log Kow 2.58-2.53 (pH 1)
10. STABILITY AND REACTIVITY	
Hazrdous reactions	: Reactive to alkaline compounds
Hazards decomposition products : Toxic and irritating hydrogen chloride or toxic gases may form in case of fire.	
11. TOXICOLOGICAL INFORMATION	
Inhalation	: Rat LC ₅₀ >1.36 (mg/L)
Dermal	: Rat LD ₅₀ > 4932 mg/kg
Oral	: Rat LD ₅₀ 500 mg/kg
Skin irritation : 2,4-D Dimethyl ammonium 84% w/v SL did not cause skin reaction. The irritancy index was 0.0 (Nil).It classifies 2,4-D Dimethyl ammonium 84% w/v SL as non irritant to skin in rabbit.	
Eye irritation : 2,4-D Dimethyl ammonium 84% w/v SL is classified as severely irritating to eye in rabbit, based on the maximum irritancy index at 48 hours.	
Allergic sensitisation : 2,4-D Dimethyl ammonium 84% w/v SL is found to be a non sensitizer in the standard guinea pig sensiti on test.	
12. ECOLOGICAL INFORMATION	
Persistence and Degradation : Half life in soil is less than 7 days.	
Toxicity to birds : Acute oral LD ₅₀ - Wild Ducks : >1000 mg/kg	
Japanese Quail : 668 mg/kg	
Pigeons : 668 mg/kg	
Pheasants : 472 mg/kg	
LC ₅₀ - Mallard Ducks : >5620 mg/l	
Toxicity to fish : LC ₅₀ (96 h) - Rain bow trout : > 100 mg/l	
LC ₅₀ (21 d) - Daphnia : 235 mg/l	
Toxicity to Algae : EC ₅₀ (5 d) - Selenastrum capricornutum : 33.2 mg/l	
Toxicity to bees : LD ₅₀ Oral - 104.5 µg / bee	
Effect on efflunt treatment : No Data	

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13. DISPOSAL CONSIDERATIONS Disposal should be in accordance with local or national legislation
14. TRANSPORT INFORMATION 2,4-D Dimethyl Ammonium 84% w/v SL, toxic, n.o.s. fall in DOT Hazard class 9 and 2,4-D Dimethyl Ammonium 84% w/v SL packing group III. The label required is “ Keep Away From Food ” .
15. REGULATORY INFORMATION 2,4-D Dimethyl Ammonium 84% w/v SL is used as Herbicide
16. OTHER INFORMATION This data sheet is prepared in accordance with directive 91/155/EEC This product should be stored, handled and used in Accordance with good industrial hygiene practices and in conformity with any legal regulations. The information contained herein is based on the present state of our knowledge and is intended to describe our product from the point of view of safety requirements. It should not, therefore, be construed as guaranteeing specific properties. Recipients of our product must take responsibility for observing existing laws and regulations.